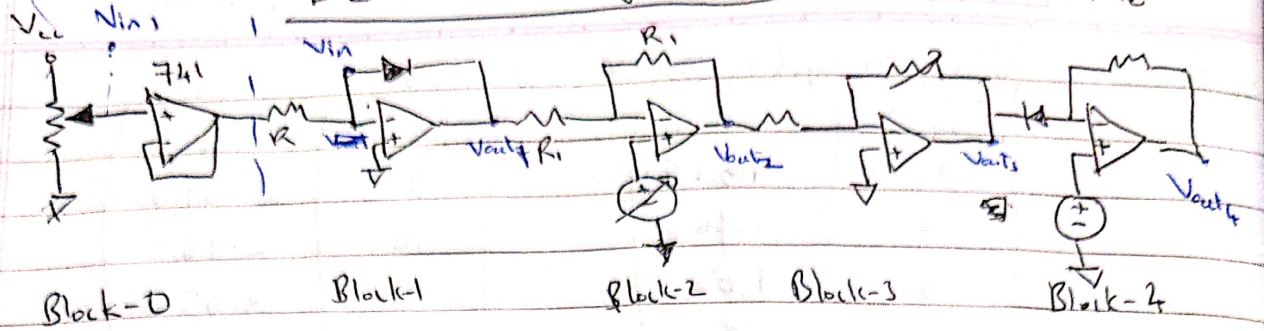
EE 230 – Analog lab - 5

Fine



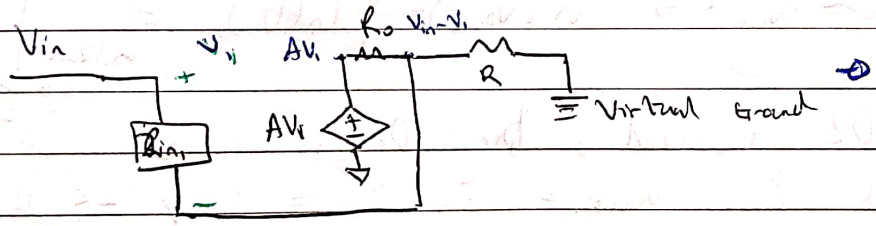
Block-0 Block-1 Block-2 Block-3 Block-4

Ans.1) Block-0 is a buffer after potentiometer so as to provide the voltage of potentiometer without its internal resistance. Also, protects potentiometer from having to supply current.

Ans-2) If V_{in}, V_{in1} directly connected ⇒ potentiometer is loaded & ~~[illegible]~~ its internal resistance ~~to~~ makes V_{in} drop from before. Also, its range now is $\left[0, \frac{R}{\frac{R_{in}}{4}+R}\right] V_{cc}$ with most drop ~~when~~ ~~$V_{in1}=V$~~ at halfway position

$(1-\theta)R_{in}$
$+R_{in}$

Ans-3)



$i_{tot} = V_1/R_{in1}$; $V_{in}\left(\frac{1}{R_o}+\frac{1}{R}\right) = V_1\left(\frac{1}{R_o}+\frac{1}{R}\right) + \frac{AV_1}{R_o} \Rightarrow V_1 = \frac{V_{in}\left(\frac{1}{R_o}+\frac{1}{R}\right)}{\frac{1}{R_o}+\frac{1}{R}-\frac{A}{R_o}}$

$\Rightarrow i_{tot} = \frac{V_{in}}{R_{in}} \frac{\left(\frac{1}{R_o}+\frac{1}{R}\right)}{\left(\frac{1}{R_o}+\frac{1}{R}+\frac{A}{R_o}\right)}$ ⇒ ~~$Z_{in} = \frac{1}{\left(\frac{1}{Z_{o1}}+\frac{1}{R}\right)}+1$~~

$$\boxed{Z_{in} = Z_{in}(A+1)\frac{\left(1+\frac{Z_o}{R}\frac{1}{A+1}\right)}{1+\frac{Z_o}{R}}}$$

We checked voltage of potentiometers V_{b1}, V_{b2} as needed

Ans.4) Final values

$R = 15K$

$R_1 = 5K$

$R_{21} = 10K$

$R_{22} = 4.78K$ $R_3 = 10K$

$V_{b1} = -0.29\ V$

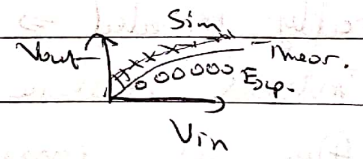
$V_{b2} = 0.61\ V$

Scanned with OKEN Scanner

Ans 8) V_{in} = sweep(1, 15) V , step = 1V

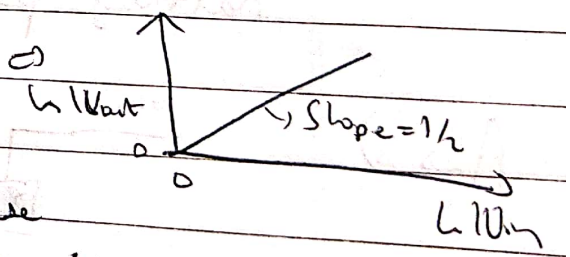
V_{in}	V_{out}	V_{in}	V_{out}
1	1.5 V	9	2.87 V
2	1.76 V	10	2.96 V
3	1.82 V	11	3.05 V
4	2.16 V	12	3.28 V
5	2.31 V	13	3.37 V
6	2.47	14	3.48 V
7	2.58	15	
8	2.78		

Ans. 9) Plot in report.



Ans. 10) Plot in report

Ans. 11) $V_{out} \approx V_{in}^{1/2} \Rightarrow \ln(V_{out}) = \frac{1}{2}\ln(V_{in})$



Ans. 12) If D2 reversed, then D_4 is reverse biased & $V_{out} = I_{S,2} R_3 \approx 1mV << 26mV$ of noise

$\therefore$ No output